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TUTORIAL

# Differential Amplifiers

*Differential Pairs, Common-Mode Response, MOS Loads, and the Gilbert Cell*

Department of Electronics and Information Engineering | Ho Chi Minh City University of Technology and Engineering (HCM-UTE), Vietnam

**Based on Design of Analog CMOS Integrated Circuits, Chapter 4 (Razavi)**

OUTLINE

# Roadmap for This Lecture

*Six sections, from differential signaling to the Gilbert cell*

## I Single-Ended vs. Differential Signals

- Why differential signaling rejects common-mode noise
- Clock-coupling and supply-noise examples

## II Basic Differential Pair

- Structure, symmetry, and input common-mode range
- Maximum output swing

## III Differential Gain — Method 1 & Half-Circuit

- Method 1: differential pair as a degenerated CS stage
- Half-circuit technique and the virtual-ground argument

## IV Common-Mode Response

- Common-mode gain and the role of tail impedance
- CMRR degradation: RD mismatch and gm mismatch

## V Differential Pair with MOS Loads

- Diode-connected loads and gain-boosting current sources
- Cascode differential pair

## VI Gilbert Cell

- Variable-gain amplifier concept
- Combining two differential pairs

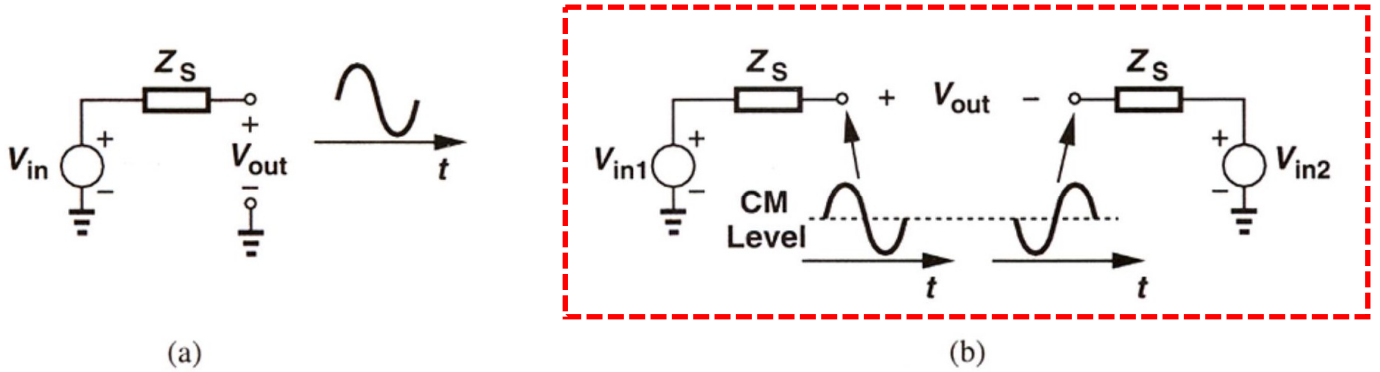
I

# Single-Ended vs. Differential Signals

- Why differential signaling rejects common-mode noise
- Clock-coupling and supply-noise examples

# Single-Ended vs. Differential Signals

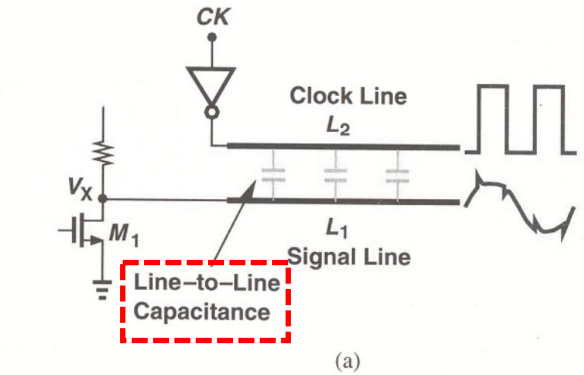
- A single-ended signal is referenced to a fixed ground; a differential signal is the difference between two complementary nodes, each swinging around a common-mode (CM) level
- Differential signaling rejects noise that couples equally into both nodes, since only the difference between them carries information




**(a) Single-ended and (b) differential signals**

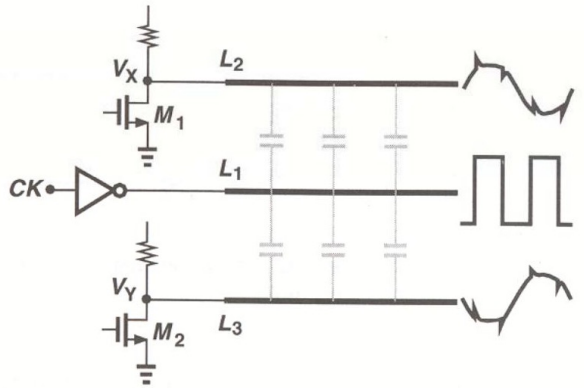
# Noise Immunity Example — Clock Coupling (*)

- A single signal line L1 running parallel to a clock line L2 picks up clock noise through line-to-line coupling capacitance
- Making the signal path differential (Vx, Vy) lets the coupled clock noise appear as a common-mode disturbance on both lines, which cancels at the differential output




(a)

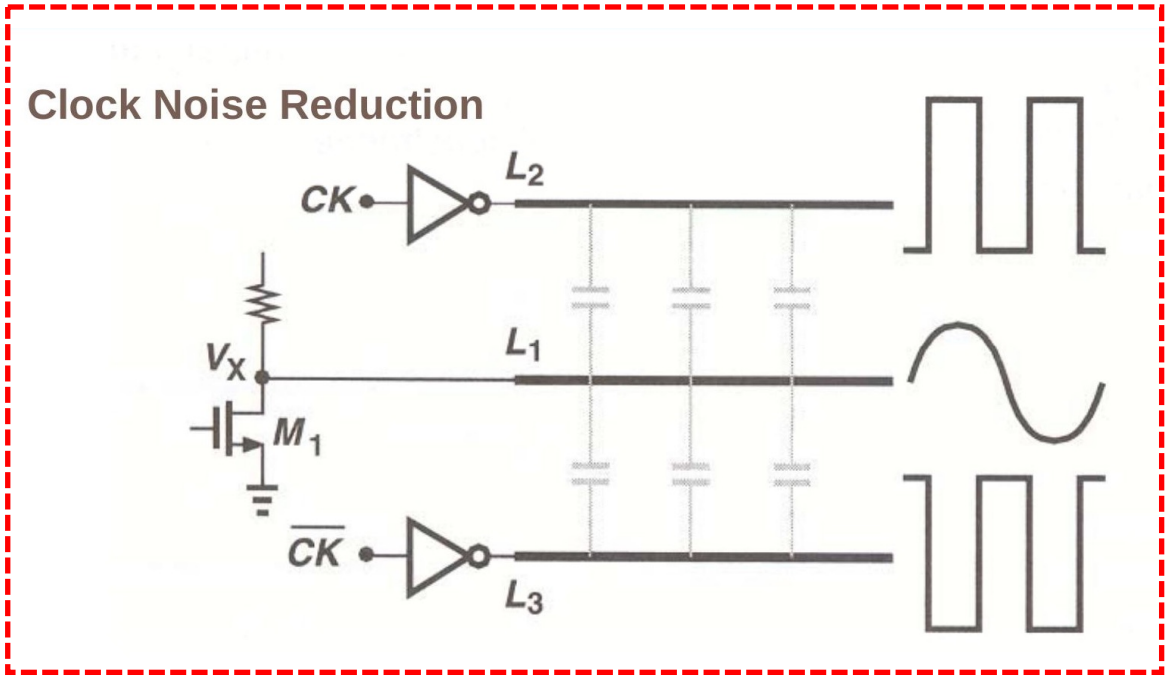
**(a) Corruption of a signal due to coupling**




**(b) Reduction of coupling by differential operation; Noise will be canceled at the differential output between Vx and Vy nodes**

# Reducing Noise by Differentiating the Clock Itself (*)

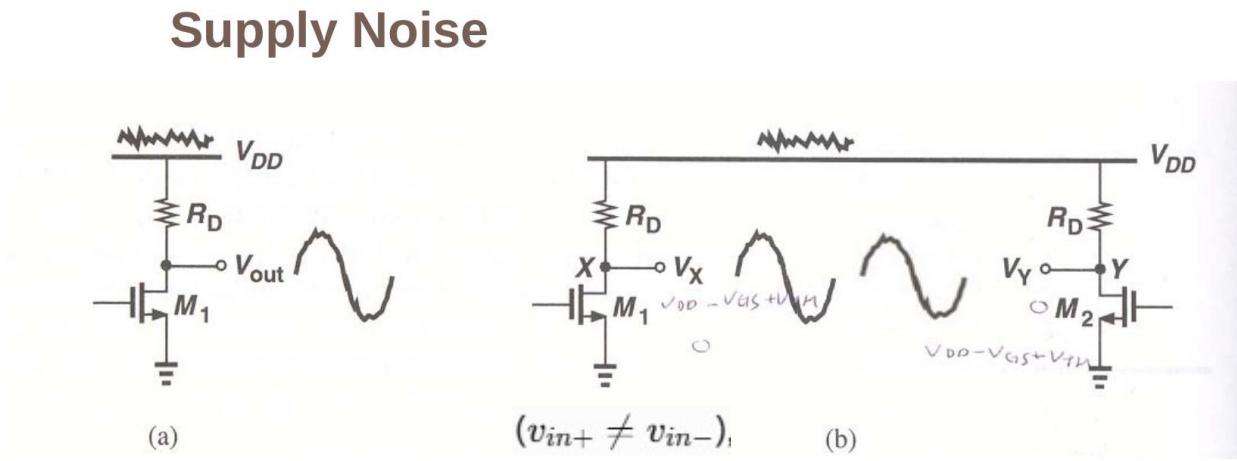
- Instead of differentiating the signal line, routing CK and its complement CK-bar on either side of the signal line L1 makes the two couplings cancel by symmetry
- This works because the two clock lines inject equal and opposite coupled charge into L1




➢**The source that is differential also eliminates noise**

# Noise Immunity Example — Supply Noise (*)

- In a single-ended CS stage, ripple on VDD couples directly into Vout through Vout = VDD − ID·RD
- In a differential pair, the same VDD ripple appears equally on both output branches and cancels in the differential output Vout1 − Vout2




➢**Supply noise on (a) single-ended circuit and (b) differential circuit**
➢**The Noise on (b) can be canceled each other**

$$I_{D1} = I_{D,bias} + g_m v_{in+}, \qquad I_{D2} = I_{D,bias} + g_m v_{in-}$$

$$V_{out} = V_X - V_Y = -g_m R_D (v_{in+} - v_{in-}) \neq 0$$

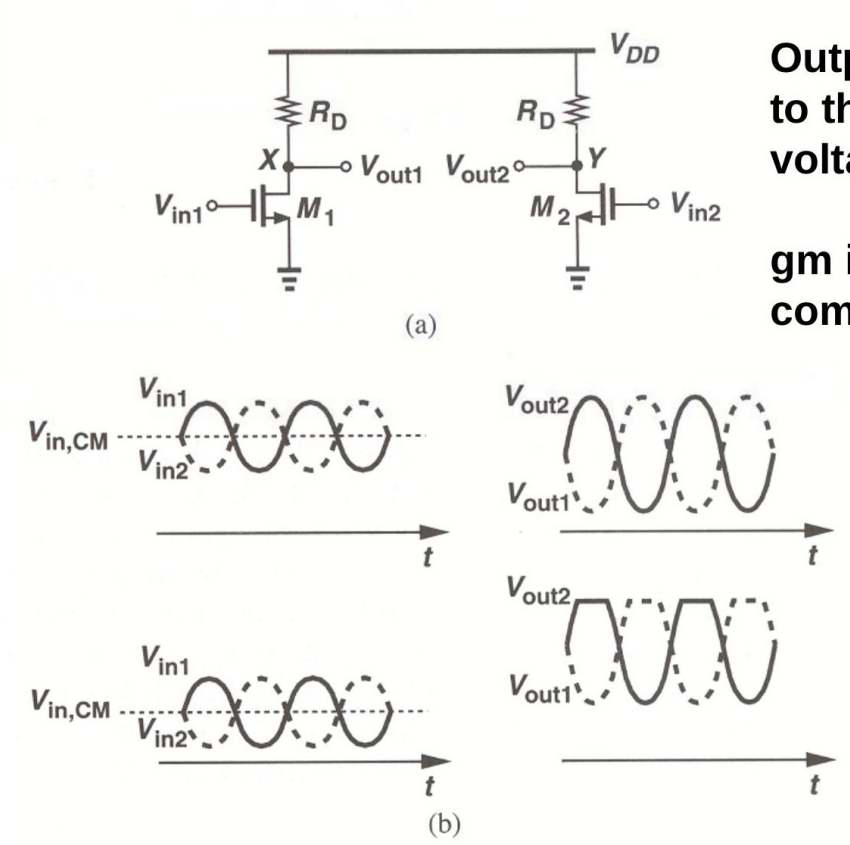
# II

# Basic Differential Pair

- Structure, symmetry, and input common-mode range
- Maximum output swing
- Effect of device width and tail current

# Why Output Depends on the Common-Mode Input

- In a simple single-transistor stage, both the output voltage and gm depend on the input common-mode (CM) level, not just the differential signal
- This CM-dependence is undesirable — the differential pair is introduced specifically to remove it

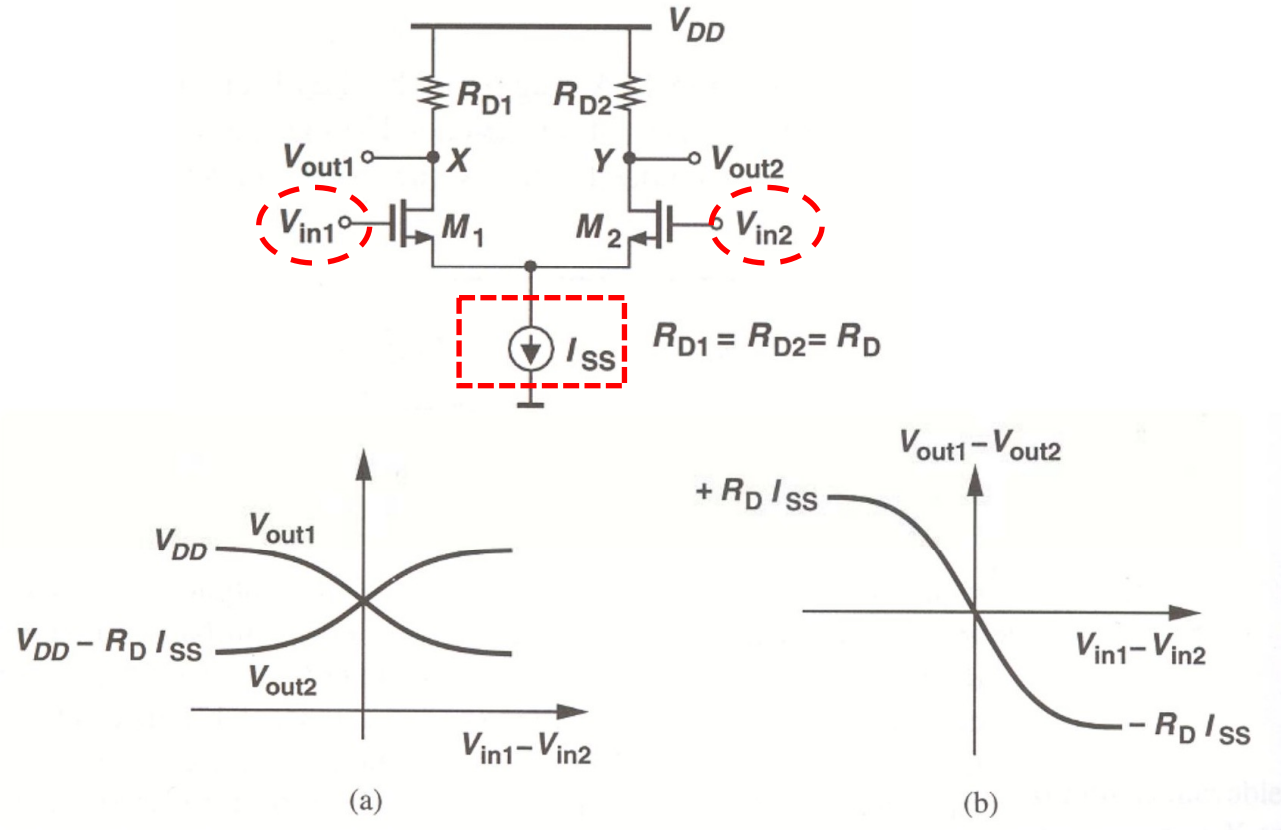



**Output voltage is very sensitive to the input common-mode voltage**

**gm is sensitive to the input common-mode voltage**

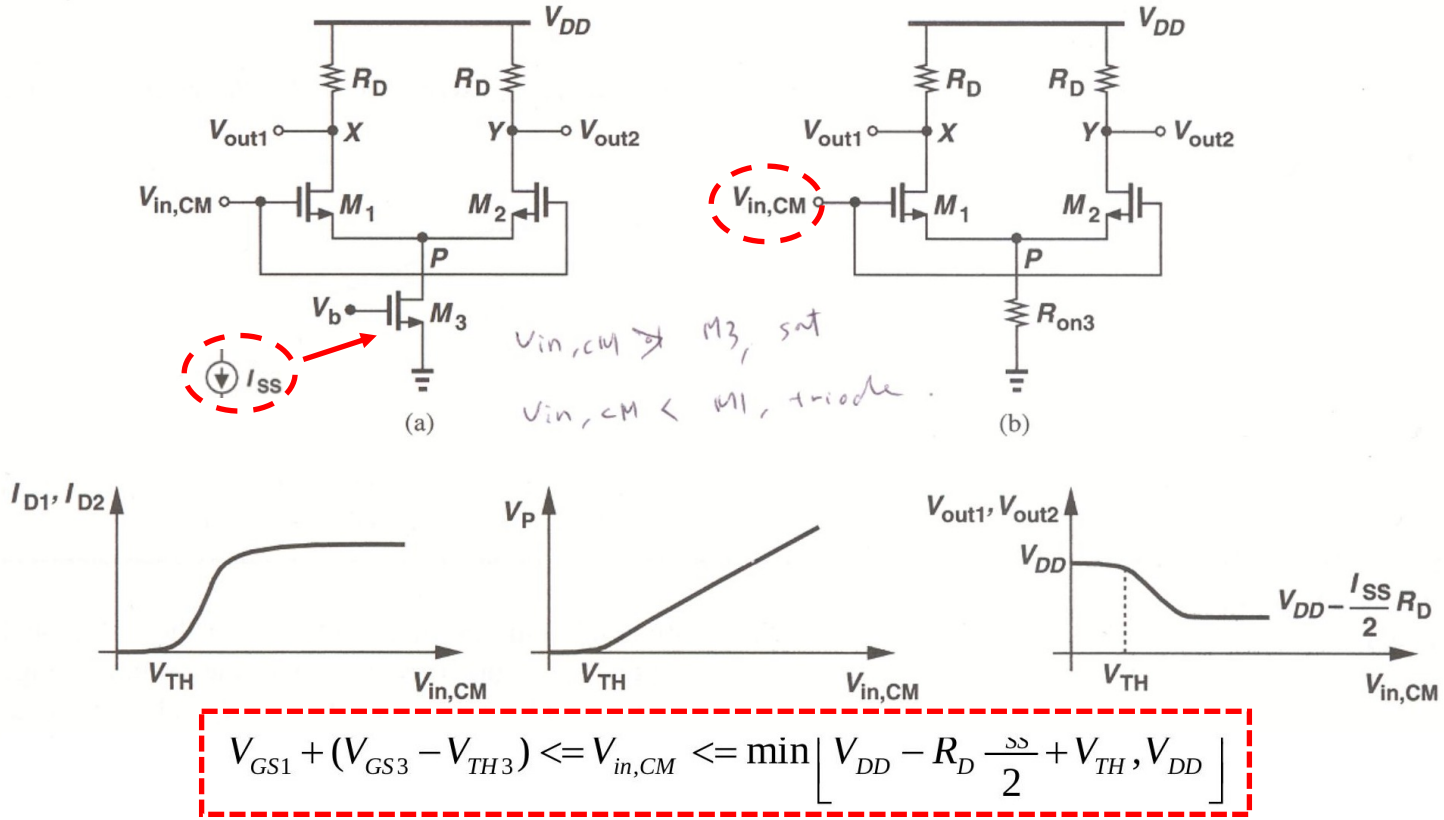
# Basic Differential Pair — Structure and Symmetry

- M1 and M2 share a common tail current source ISS, with matched loads RD1 = RD2 = RD
- When Vin1 = Vin2, the circuit is fully symmetric: ISS splits evenly and Vout1 = Vout2
- As Vin1 − Vin2 sweeps, the tail current steers between M1 and M2, producing the classic differential transfer characteristic



# Input Common-Mode Range

- **For both transistors to stay in saturation and the tail current source to remain in compliance**, Vin,CM must satisfy VGS1 + (VGS3−VTH1) ≤ Vin,CM ≤ min[VDD−RD·ISS/2+VTH, VDD]
- Operating outside this range collapses the tail current source or pushes a transistor into triode, distorting the differential response

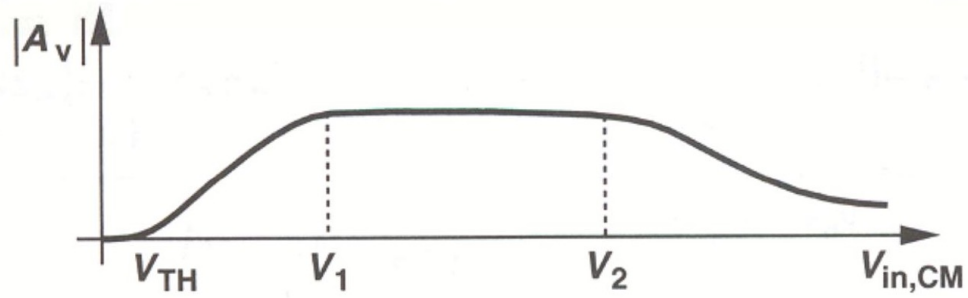



$$V_{GS1} + (V_{GS3} - V_{TH3}) <= V_{in,CM} <= \min\left[V_{DD} - R_D \frac{I_{SS}}{2} + V_{TH}, V_{DD}\right]$$

# Ex. 4.1 — Sketching |Av| vs. Input CM Level

- The small-signal differential gain |Av| stays flat and maximal over the valid input CM range where both devices remain in saturation
- Outside the valid range (below VTH or beyond the swing limits V1, V2), |Av| rolls off as devices leave saturation or turn off

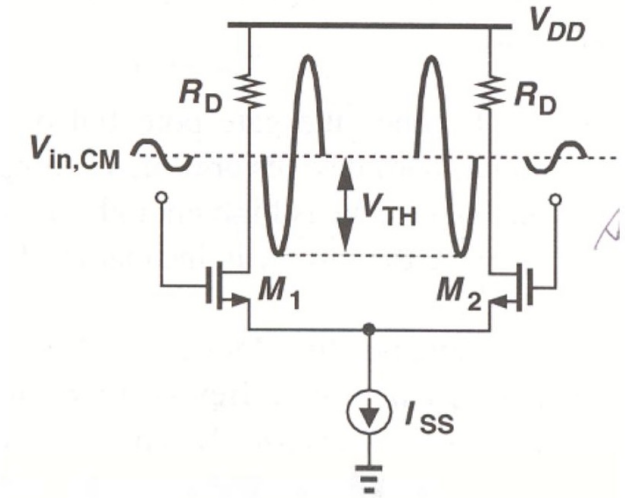
Example4.1
Sketch the small-signal differential gain of a differential pair as a function of the input CM level



# Maximum Allowable Output Swing

- Each output node must stay below VDD and above the point where its transistor enters triode
- The valid single-ended output range is Vin,CM − VTH < Vout < VDD, setting the ceiling on achievable voltage swing

Maximum allowable output swing in a differential pair




$$V_{in,CM} - V_{TH} < V_{out} < V_{DD}$$

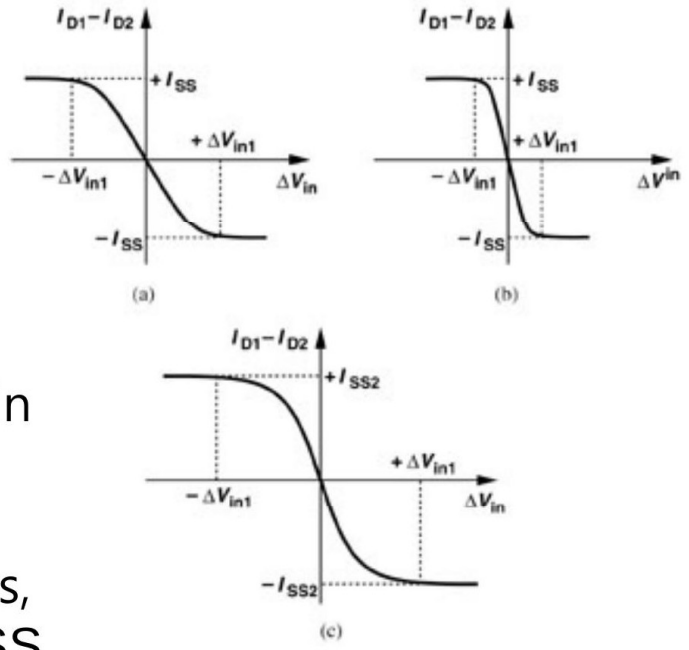
# Ex. 4.2 — Effect of Device Width and Tail Current

- As device width W increases, the differential pair's transition region ΔVin (the input range over which current fully steers) becomes narrower — the pair switches more abruptly
- As the tail current ISS increases, ΔVin becomes wider and the maximum steerable current ISS itself increases

Ex4.2)
Plot the input-output characteristic of a differential pair as the device width and the tail current vary

As the width increases, ΔVin becomes narrower

As the tail current increases, ΔVin becomes larger and ISS becomes larger



III

# Differential Gain — Method 1 & Half-Circuit

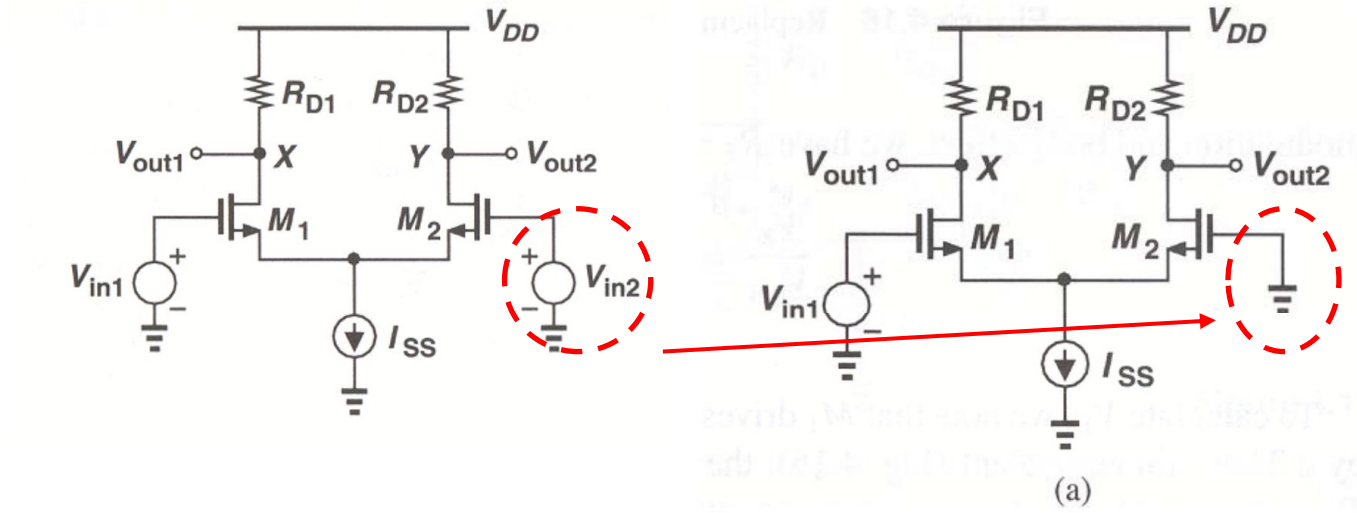
- Method 1: differential pair as a degenerated CS stage
- Half-circuit technique and the virtual-ground argument
- Worked examples with matched and mismatched devices

# How to Get Voltage Gain? — Method 1 Setup (*)

- Method 1 treats the differential pair as a common-source stage with source degeneration, where M2 provides the degeneration impedance seen from M1's source
- With Vin2 held at AC ground, the impedance looking into M2's source is approximately 1/gm2

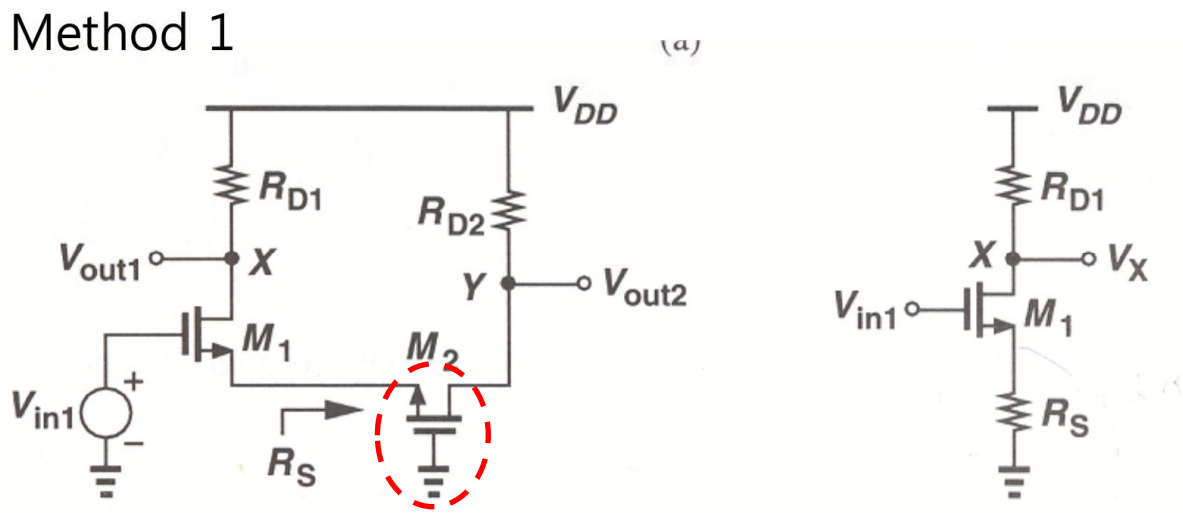
How to get voltage gain?

Method 1



# Method 1 — Completing the Gain Calculation

- Using **RS = 1/gm2**, the degenerated common-source formula gives Vout1/Vin1 = −RD/(1/gm1+RS)
- **Assuming gm1 = gm2 = gm** (matched devices), this simplifies to Vout1/Vin1 = −gm·RD/(1+gmRS) = −(gm/2)·RD




$$R_s = 1/g_{m2}$$

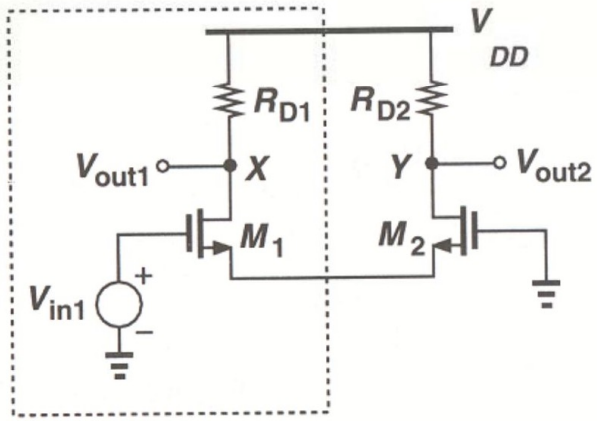
$$V_{out1}/V_{in1} = -R_D/(1/g_{m1} + R_s) = -\frac{g_m R_D}{1+g_m R_S} = -\frac{g_m}{2} R_D$$

# Diff Pair as a CS, CD, and CG Amplifier

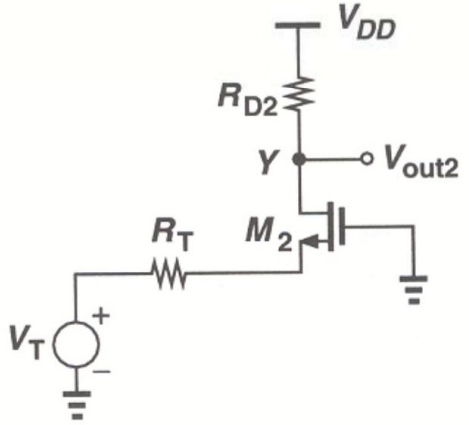
- The differential pair can be viewed as M1 in common-source configuration cascaded with M2 acting as a common-gate stage, connected through a common-drain (source-follower)-like node
- This composite view explains both the direct gain path (through M1) and the cross-coupled gain path (through M2 acting as CG)

Method 1

Diff Pair as a CS, CD, and CG Amp




(a)




(b)

# Cross-Gain: Vin1 to Vout2

- Looking from M2's source, M1 presents an equivalent Thévenin resistance RT = 1/gm1
- Treating M2 as a common-gate stage driven by this equivalent source gives Vout2/Vin1 = RD/(1/gm2+RT) = (gm/2)·RD — same magnitude, opposite sign from the direct path

Method 1

Diff Pair as a CS, CD, and CG Amp

$$R_T = 1/g_{m1}$$

$$V_{out2}/V_{in1} = V_{out2}/V_T = R_D/(1/g_{m2} + R_T)$$

$$= \frac{g_m}{2} R_D$$

# Superposition — Total Differential Gain (Matched Case) (**)

- (Vout1−Vout2) due to Vin1 alone = −(gm/1)·RD·Vin1 — the two contributions add because they have opposite signs at Vout1 and Vout2
- By symmetry, (Vout1−Vout2) due to Vin2 alone gives the same coefficient, so the total differential gain is (Vout1−Vout2)/(Vin1−Vin2) = −gm·RD

Method 1

Diff Pair as a CS, CD, and CG Amp

$$(V_{out1} - V_{out2}) \text{ due to } V_{in1} = -\frac{g_m}{1} R_D V_{in1}$$

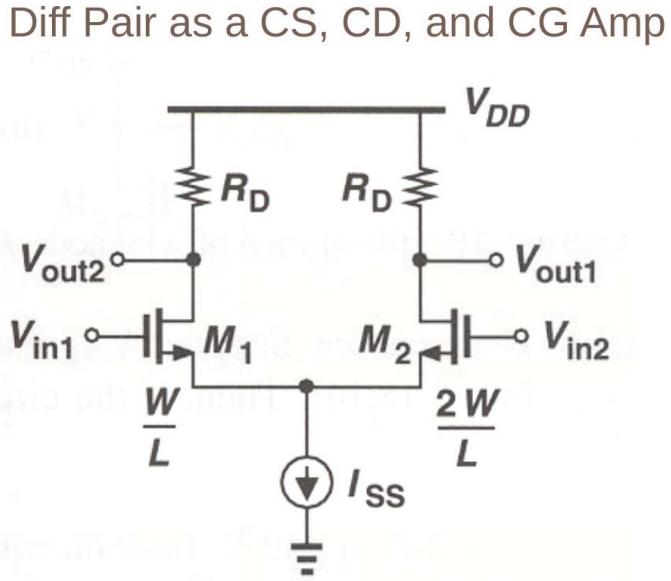
$$(V_{out2} - V_{out1}) \text{ due to } V_{in2} = -\frac{g_m}{1} R_D V_{in2}$$

$$(V_{out1} - V_{out2}) \text{ total} = -\frac{g_m}{1} R_D (V_{in1} - V_{in2})$$

# Ex. 4.3 — Gain with Mismatched Device Widths

- Setup: M2's width is twice M1's (W/L ratio 2:1), so gm2 = 2·gm1 while Vin1 and Vin2 share the same DC bias
- This breaks the perfect symmetry assumed in the matched-device case, requiring the general Method 1 / half-circuit analysis to find the actual gain

**Ex4.3) calculate the small signal gain if the bias of Vin1 and Vin2 are equal**

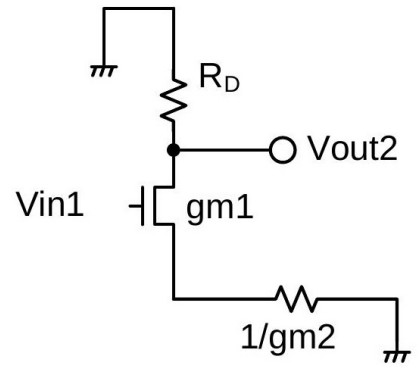


# 4.2.2 — Applying the Half-Circuit Idea (gm1 ≠ gm2)

- Redrawing the Vin1-to-Vout2 path as a CG stage with source degeneration 1/gm2, the gain is Vout2/Vin1 = −Gm·Rout
- This generalizes the Method-1 formula from slide 15 to the case where gm1 and gm2 differ

**4.2.2 quantitative analysis**

Diff Pair as a CS, CD, and CG Amp



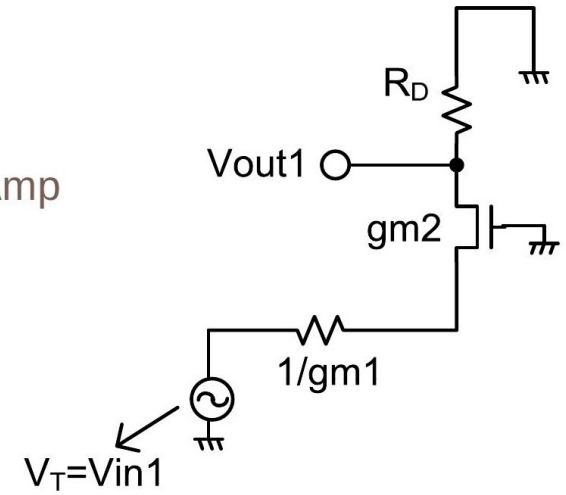

$$\frac{V_{out2}}{V_{in1}} = -Gm \cdot Rout = -\frac{g_{m1} \cdot R_D}{1+\frac{g_{m1}}{g_{m2}}} = -\frac{g_{m1} \cdot R_D}{1+\frac{1}{2}} = -\frac{2}{3} g_{m1} R_D$$

# Completing Example 4.3 — the Vout1/Vin1 Path

- By symmetry with the previous slide (swapping the roles of gm1 and gm2), Vout1/Vin1 = −Gm·Rout = (2/3)·gm1·RD
- This uses gm2 = 2·gm1 from the problem's device-sizing assumption

**4.2.2 quantitative analysis**

Diff Pair as a CS, CD, and CG Amp




$$\frac{V_{out1}}{V_{in1}} = -Gm \cdot Rout = \frac{g_{m2} \cdot R_D}{1+\frac{g_{m2}}{g_{m1}}} = \frac{2g_{m1} \cdot R_D}{1+2} = \frac{2}{3} g_{m1} R_D$$

# Ex. 4.3 — Final Differential Gain

- Combining both paths: Av = (Vout2−Vout1)/Vin1 = −(4/3)·gm1·RD
- Since Vin2 is held fixed, this also equals the standard differential gain (Vout2−Vout1)/(Vin1−Vin2) = −(4/3)·gm1·RD — larger in magnitude than the matched-device result of −gm·RD

**4.2.2 quantitative analysis**

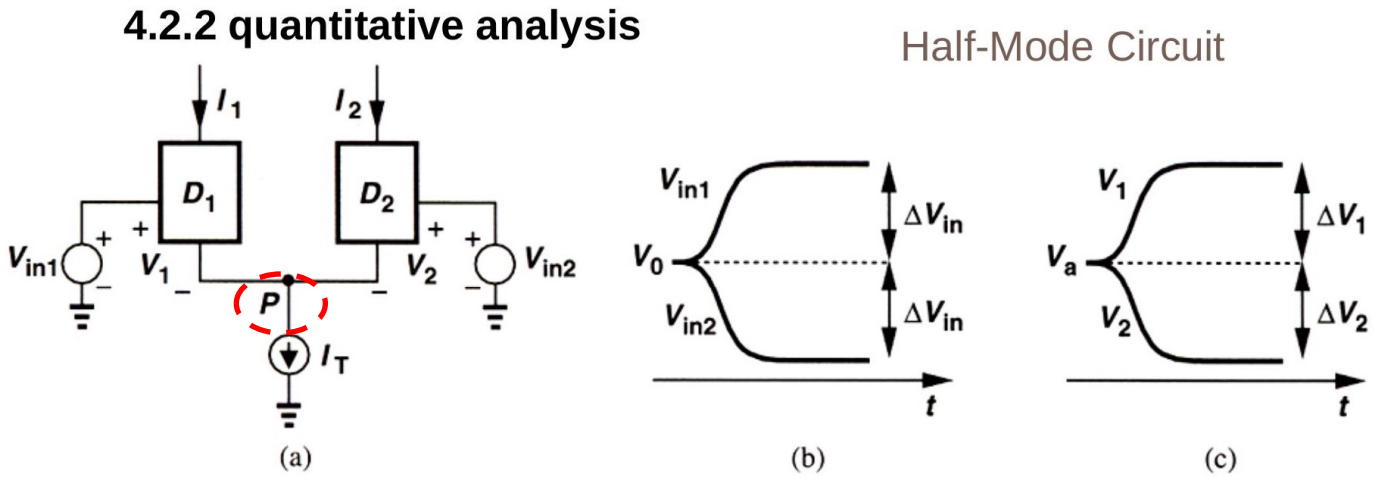
Diff Pair as a CS, CD, and CG Amp

$$\therefore A_V = \frac{V_{out2} - V_{out1}}{V_{in1}} = -\frac{4}{3} g_{m1} R_D$$

$$\therefore A_V = \frac{V_{out2} - V_{out1}}{V_{in1} - V_{in2}} = -\frac{4}{3} g_{m1} R_D$$

# Half-Circuit Technique — the Virtual-Ground Argument (*)

- For two identical nonlinear devices D1, D2 sharing a tail node P and driven by perfectly differential inputs (+ΔV, −ΔV), node P does not move — it is a virtual ground
- Because the tail current IT is fixed, symmetry forces the P-node voltage to stay constant under purely differential excitation



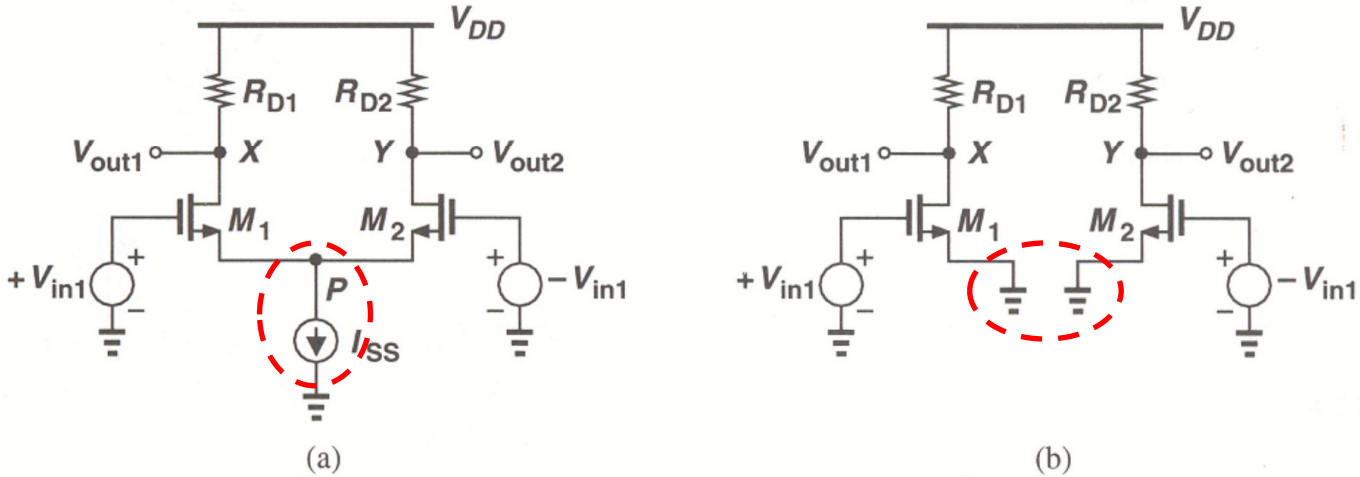

➢**Node P: virtual ground**
➢**$I_T$ is constant => Voltage on P node not to change**

# Applying the Half-Circuit to the Standard Diff Pair

- Under pure differential excitation, the tail node P is a virtual ground, so each half of the circuit reduces to a single common-source stage with its source tied to true ground
- This directly gives |Vout1/Vin1| = gm1RD1 and |Vout2/Vin2| = gm2RD2 — a major simplification over full node-equation analysis

**4.2.2 quantitative analysis**

Half-Mode Circuit



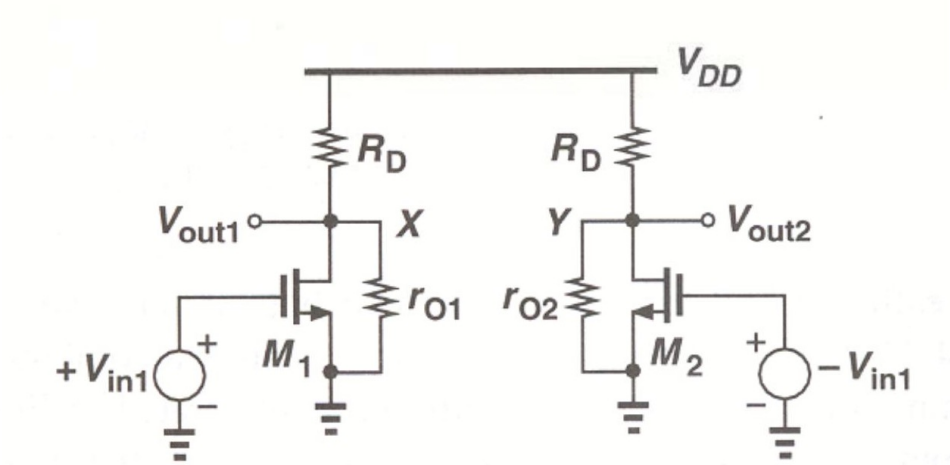

$$\left|\frac{V_{out1}}{V_{in1}}\right| = g_{m1}R_{D1}, \left|\frac{V_{out2}}{V_{in2}}\right| = g_{m2}R_{D2} \Rightarrow \left|\frac{V_{out1}-V_{out2}}{V_{in1}-V_{in2}}\right| = g_m R_D$$

# Ex. 4.4 — Differential Gain via the Half-Circuit

- Using the CS half-circuit that includes finite output resistance ro, each side contributes −gm(RD∥ro)
- The full differential gain is (Vx−Vy)/(2Vin) = −gm(RD∥ro), matching the standard CS-stage gain applied to each half-circuit

**Ex4.4) calculate the differential gain**

Half-Mode Circuit



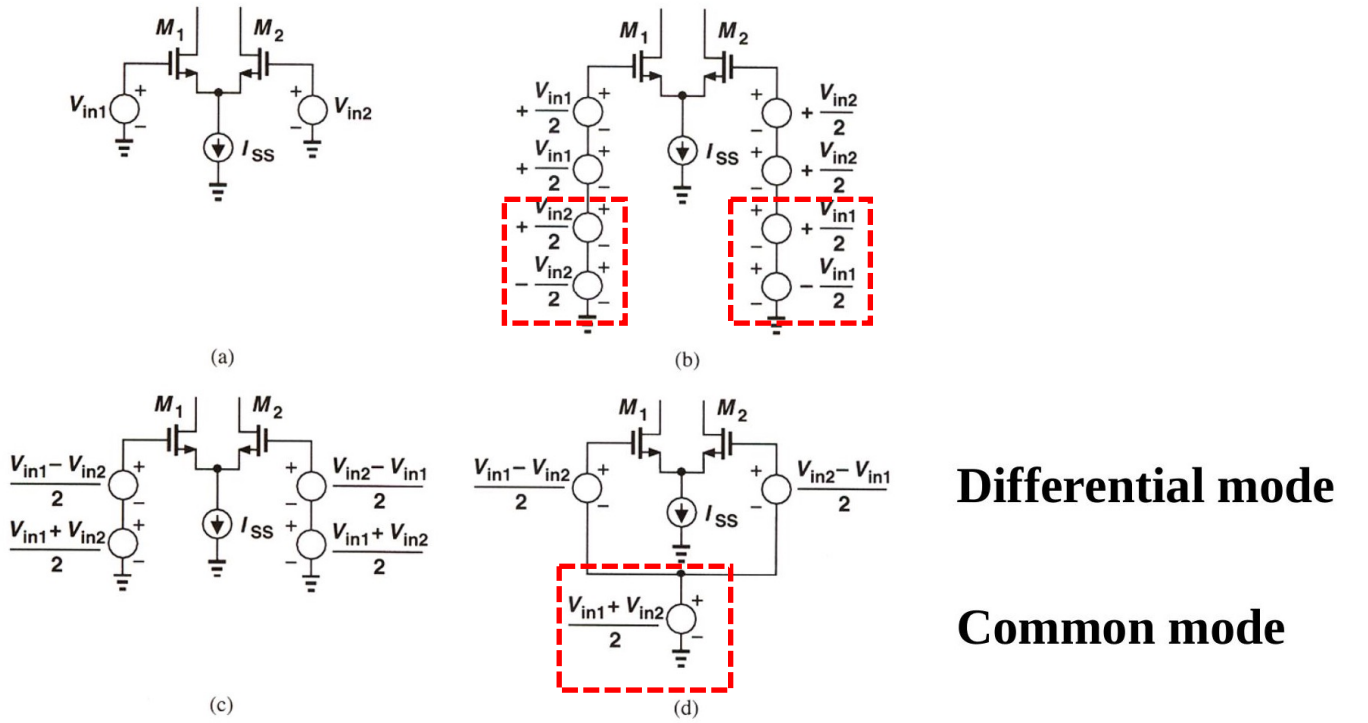

$$\frac{V_X - V_Y}{2V_{in}} = -gm\left(R_D \parallel r_o\right)$$

# Decomposing Any Input Pair into DM and CM Components

- Any input pair (Vin1, Vin2), even asymmetric ones, can be decomposed into a differential-mode component ±(Vin1−Vin2)/2 and a common-mode component (Vin1+Vin2)/2
- This decomposition lets superposition be applied: analyze the DM and CM responses separately, then add them

**4.2.2 quantitative analysis**

➢**Vin1 not equal to Vin2**




**Differential mode**

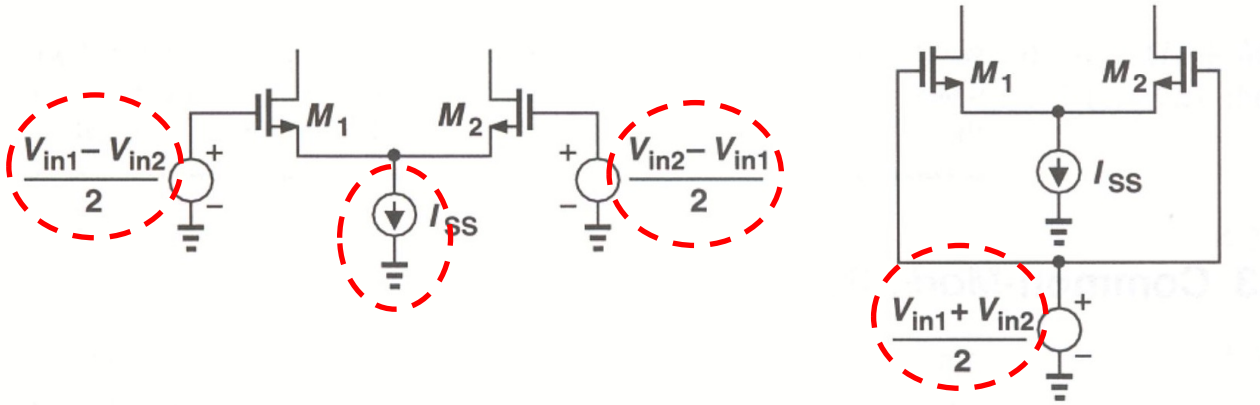
**Common mode**

# Superposition of DM and CM Half-Circuits

- The differential-mode half-circuit uses ±(Vin1−Vin2)/2 at each gate with the tail node treated as a true ground (virtual ground under DM excitation)
- The common-mode half-circuit uses (Vin1+Vin2)/2 at both gates with the tail impedance retained explicitly, since it is not a virtual ground under CM excitation

**4.2.2 quantitative analysis**

➢Superposition of differential and common-mode signals

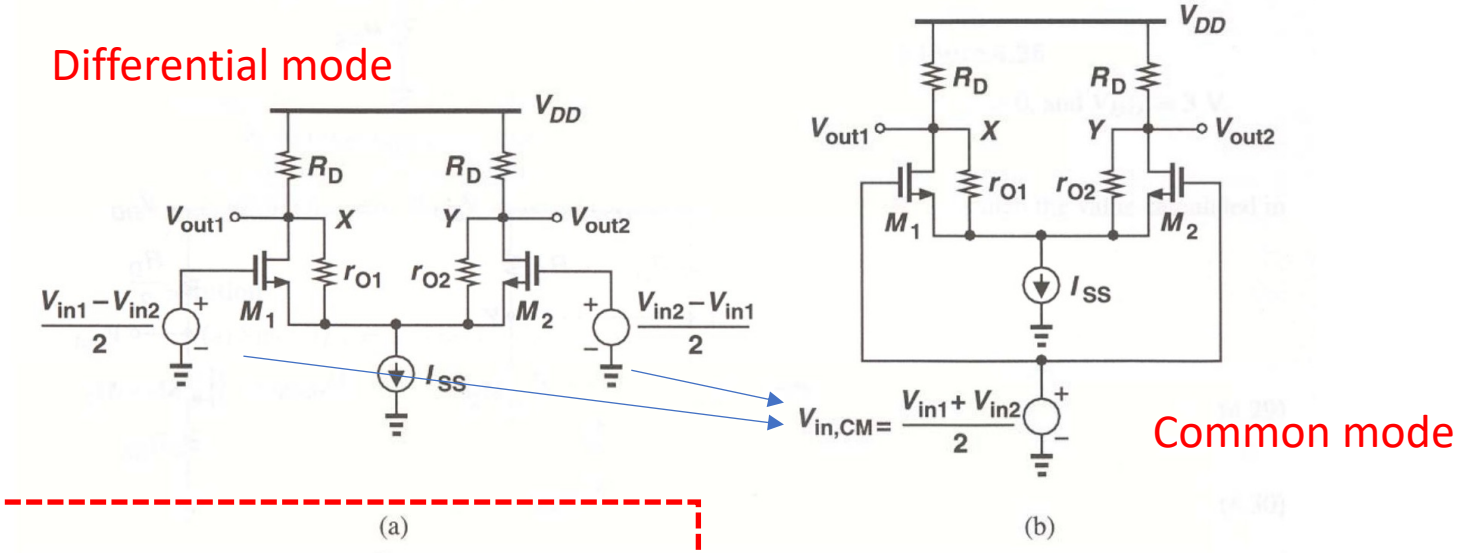


# Ex. 4.5 — Differential and Common-Mode Gain Together

- The differential-mode half-circuit (including ro) directly gives VX−VY = −gm(RD∥ro)(Vin1−Vin2), inherited from the Ex. 4.4 result
- The common-mode half-circuit requires the tail resistance to be included explicitly — analyzed on the following slides

Ex4.5) calculate the differential gain and common mode gain

➢Superposition of differential and common-mode signals




$$V_X - V_Y = -g_m(R_D \parallel r_o)(V_{in1} - V_{in2})$$

# Bridge Question — CM Gain with an Ideal Tail Source

- What is the common-mode gain in Ex. 4.5 if the tail current source is ideal (infinite output resistance)?
- This question motivates the detailed common-mode response analysis that follows

What is common-model gain in ex4.5 when the tail current source is ideal?

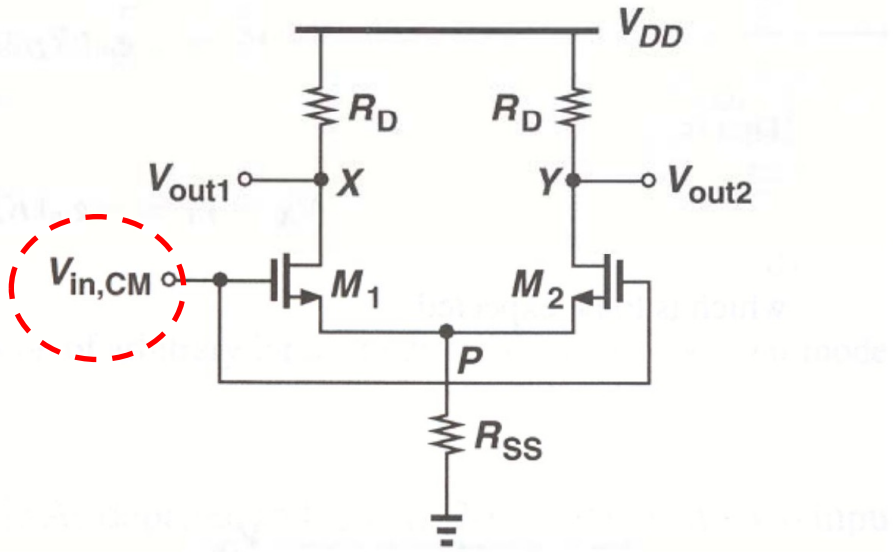
# IV

# Common-Mode Response

- Common-mode gain and the role of tail impedance
- CMRR degradation: RD mismatch and gm mismatch
- Frequency dependence and cascaded stages

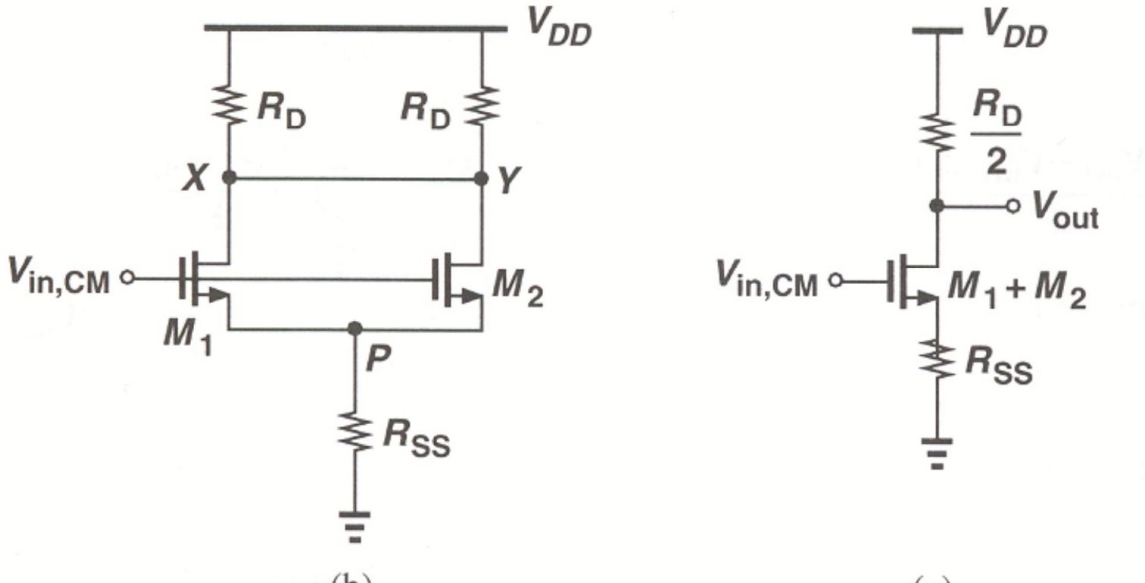
# Common-Mode Response — Circuit Setup

- Under common-mode excitation, both gates move together (Vin,CM) while the tail node P now carries a finite impedance RSS to ground
- Because P is not a virtual ground here, the circuit cannot be split into independent half-circuits without keeping RSS explicit



# Common-Mode Gain — Quantitative Derivation

- Using the half-circuit with the tail impedance doubled (since each half carries only half the tail current change), Av = Vout/Vin,CM = −(RD/2) / [1/(2gm)+RSS]
- As RSS → ∞ (ideal tail source), this gain approaches zero — confirming the intuition from the bridge question




$$A_v = \frac{V_{out}}{V_{in,CM}} = -\frac{R_D/2}{1/(2g_m) + R_{SS}}$$

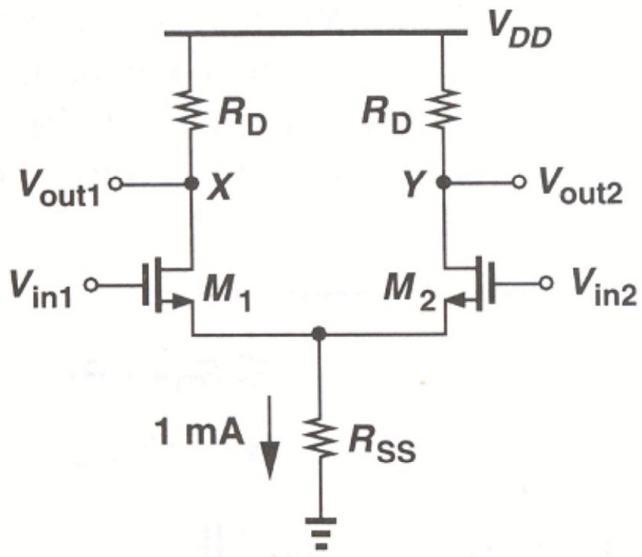
# Ex. 4.6 — Numerical Common-Mode Design Problem (**)

- Given W/L, μnCox, VTH, and VDD, find: (a) the RSS required for a given VDS margin, (b) RD for a target differential gain of 5, and (c) the output shift for a 50 mV CM step
- This ties the common-mode formulas directly to a concrete bias-design exercise

**Example4.6)**

**W/L=20/0.5, uCox=50uA/V$^2$, VTH=0.6V, and VDD=3V**

**(a) What is the required CM for which RSS sustains 0.5V?**
**(b) Calculated RD for a differential gain of 5.**
**(c) What happens at the output if the input CM level is 50mV higher than the value calculated in (a)?**



# Ex. 4.6 — Worked Solution (**)

- (a) From VGS1 = VGS2 = √(2ID1/(μnCoxW/L)) + VTH = 1.23 V, the required Vin,CM = VGS1+0.5 = 1.73 V sets the RSS voltage headroom
- (b) gm = √(2μnCox(W/L)ISS) gives RD = 3.16 kΩ for |Av| = 5; (c) a 50 mV CM step produces ΔVX,Y ≈ 96.8 mV at the output

**Example4.6)**

$$(a)\quad V_{GS1} = V_{GS2} = \sqrt{\frac{2I_{D1}}{uCoxW/L}} + V_{TH} = 1.23V$$

$$\therefore V_{in,CM} = V_{GS1} + 0.5V = 1.73V$$

$$(b)\quad gm = \sqrt{2uCox(W/L)I_{D1}} = 1/632\ \Omega$$

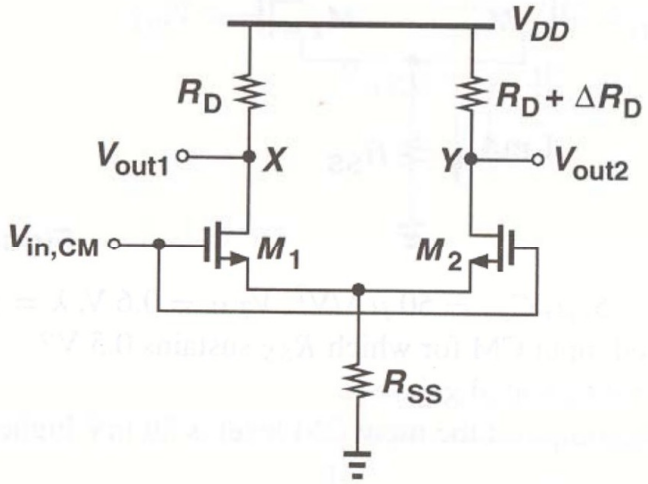
$$\therefore R_D = 3.16\ kOhm \quad to\ satisfy\ |Av| = 5$$

$$(c)\quad |\Delta V_{X,Y}| = \Delta V_{in,CM}\frac{R_D/2}{R_{SS} + 1/2gm} = 50mV \times 1.94 = 96.8mV$$

# CMRR Degradation Mechanism 1 — Load Resistor Mismatch

- Even with perfectly matched gm1 = gm2, a mismatch ΔRD between the two load resistors converts common-mode input variation into a differential output error
- ΔVX/ΔVin,CM = −gm·RD/(1+2gmRSS) on one branch and −gm(RD+ΔRD)/(1+2gmRSS) on the other — the difference is the unwanted CM-to-DM leakage

➢ $R_D$ mismatch

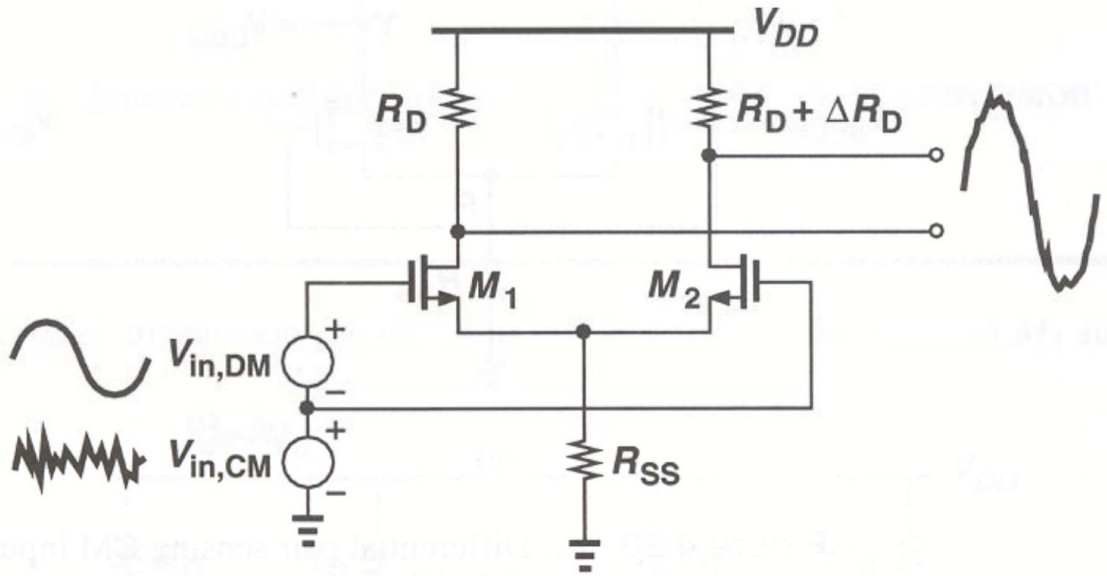



$$\frac{\Delta V_X}{\Delta V_{in,CM}} = -\frac{g_m}{1+2g_m R_{SS}} R_D, \quad \frac{\Delta V_Y}{\Delta V_{in,CM}} = -\frac{g_m}{1+2g_m R_{SS}} (R_D + \Delta R_D)$$

# Practical Consequence of CM-to-DM Conversion

- A clean differential signal Vin,DM combined with common-mode noise Vin,CM drives both gates together
- With mismatched RD, the CM noise no longer cancels perfectly and leaks into the differential output as an apparent signal — degrading CMRR in practice

Effect of CM noise in the presence of resistor mismatch



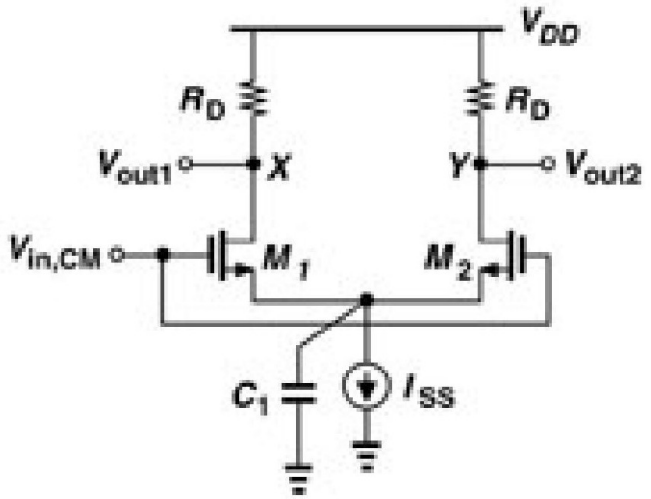
# Frequency Dependence via Finite Tail Capacitance

- A parasitic capacitance C1 at the tail node (from the tail current source and layout) is negligible at low frequency, where RSS dominates
- At higher frequency, C1's shunting admittance grows, effectively lowering the tail impedance — which increases the common-mode gain at high frequency

CM response with finite tail capacitance

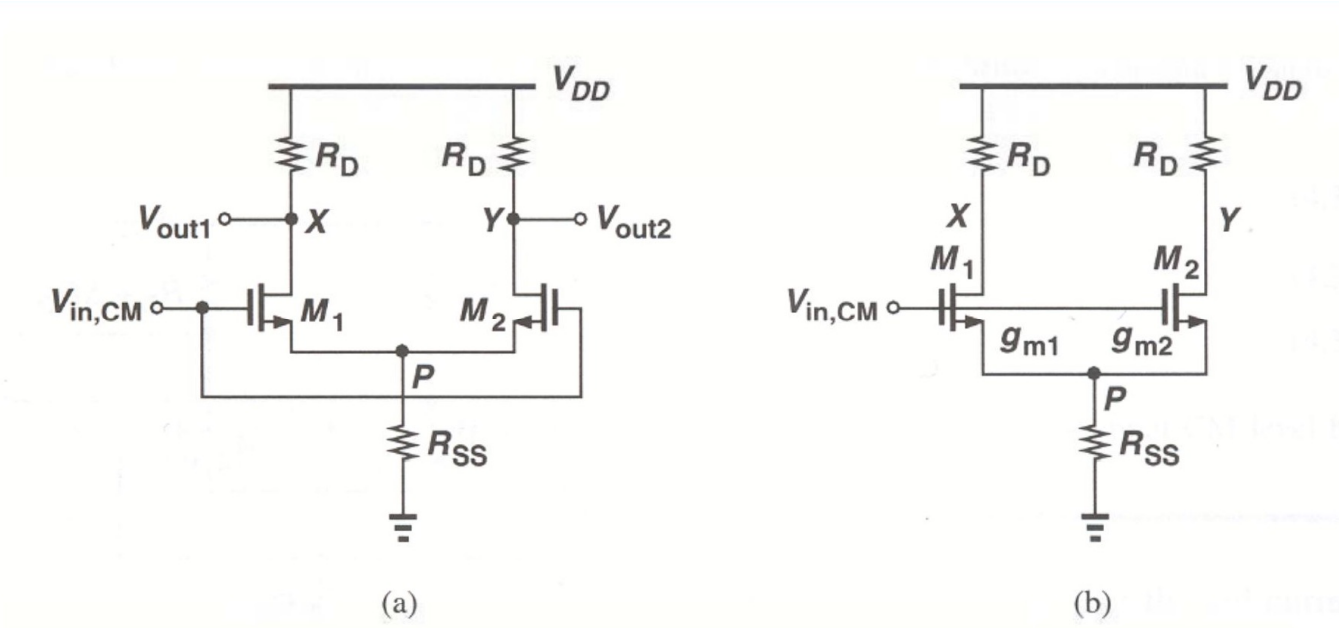
As frequency becomes higher, shunt capacitor, C1 has larger conductance

As a result, common-mode gain increases in high frequencies



# CMRR Degradation Mechanism 2 — gm Mismatch

- A second, independent cause of CMRR degradation is a mismatch between gm1 and gm2 of the input devices themselves, even with perfectly matched loads
- This mechanism is analyzed the same way — as a perturbation to the common-mode half-circuit result




➢ $g_m$ mismatch

# Full Algebra for the gm-Mismatch Case

- Small-signal drain currents: ID1 = gm1(Vin,CM−VP), ID2 = gm2(Vin,CM−VP), with KCL at the tail node giving VP = (gm1+gm2)RSS·Vin,CM / [(gm1+gm2)RSS+1]
- Substituting back gives VX = −gm1RD·Vin,CM/[(gm1+gm2)RSS+1] and the analogous VY with gm2

$$I_{D1} = g_{m1}(V_{in,CM} - V_P),\ I_{D2} = g_{m2}(V_{in,CM} - V_P)$$

$$(g_{m1} + g_{m2})(V_{in,CM} - V_P)R_{SS} = V_P$$

$$V_P = \frac{(g_{m1} + g_{m2})R_{SS}}{(g_{m1} + g_{m2})R_{SS} + 1} V_{in,CM}$$

$$V_X = -g_{m1}(V_{in,CM} - V_P)R_D = \frac{-g_{m1}}{(g_{m1} + g_{m2})R_{SS} + 1} R_D V_{in,CM}$$

$$V_Y = -g_{m2}(V_{in,CM} - V_P)R_D = \frac{-g_{m2}}{(g_{m1} + g_{m2})R_{SS} + 1} R_D V_{in,CM}$$

# CMRR with gm Mismatch — Final Formula

- VX−VY = −[(gm1−gm2)/((gm1+gm2)RSS+1)]·RD·Vin,CM, so ACM−DM = −Δgm·RD/(1+2gmRSS) with Δgm = gm1−gm2
- CMRR = |ADM|/|ACM−DM| = (gm/Δgm)·(1+2gmRSS) — larger tail impedance RSS directly improves CMRR against gm mismatch

$$V_X - V_Y = -\frac{g_{m1} - g_{m2}}{(g_{m1} + g_{m2})R_{SS} + 1} R_D V_{in,CM}$$

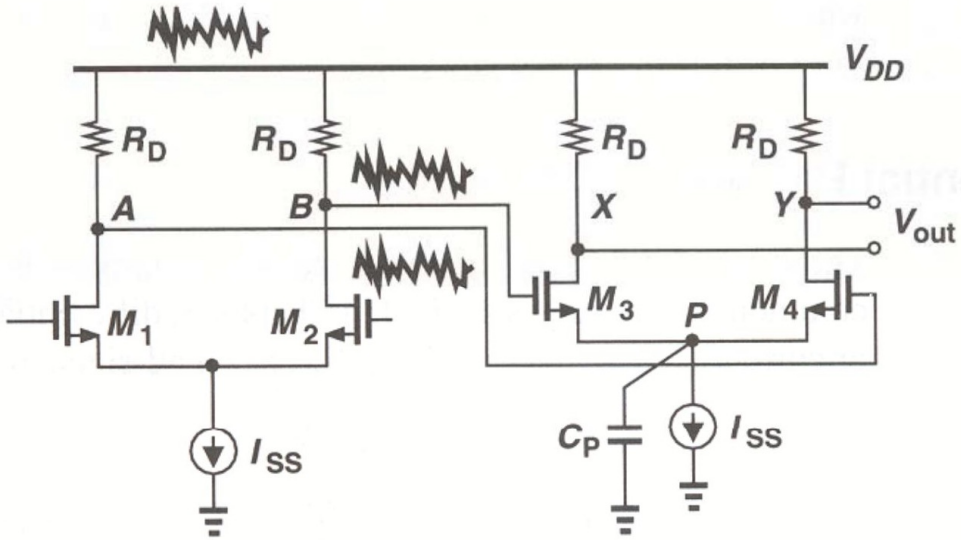
$$A_{CM-DM} = -\frac{\Delta g_m}{(g_{m1} + g_{m2})R_{SS} + 1} R_D$$

$$CMRR = \left|\frac{A_{DM}}{A_{CM-DM}}\right| = \frac{g_m R_D}{\frac{\Delta g_m R_D}{(g_{m1} + g_{m2})R_{SS} + 1}} = \frac{g_m}{\Delta g_m}(1 + 2g_m R_{SS})$$

# Ex. 4.7 — CMRR in a Cascaded Two-Stage Structure

- A first differential stage (M1, M2) drives a second stage (M3, M4) that itself suffers a gm mismatch, with a parasitic tail capacitance Cp at the second stage's tail node
- The overall $|ACM-DM| = \Delta gmRD / \sqrt{[1+(gm3+gm4)^2/(CpW)^2]}$-type expression shows the frequency-dependent degradation compounding across cascaded stages

Ex4.7) M3 and M4 suffer from a gm mismatch




$$|A_{CM-DM}| = \frac{\Delta g_m R_D}{\sqrt{1+(g_{m3}+g_{m4})^2 \left|\frac{1}{C_p w}\right|^2}}$$

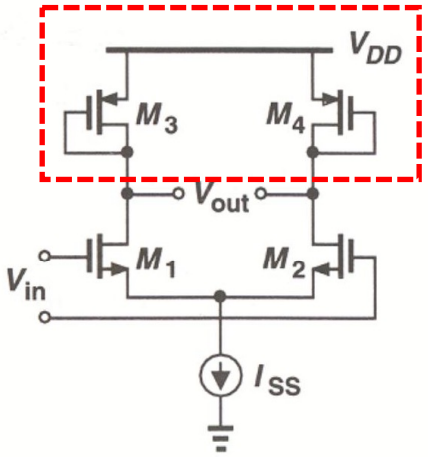
V

# Differential Pair with MOS Loads

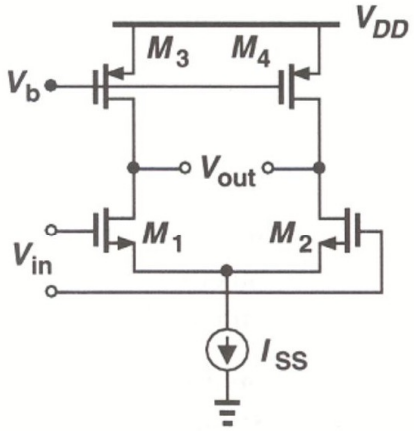
- Diode-connected loads and gain-boosting current sources
- Cascode differential pair

# Differential Pair with MOS Diode-Connected Loads

- Replacing resistive loads RD with diode-connected PMOS devices (M3, M4) removes the need for a separate bias resistor and sets the gain via a transistor ratio
- Basic gain: Av = −gmN(gmP∥roN∥roP) ≈ −gmN/gmP — a fully transistor-ratio-based, resistor-free gain expression



(a)



(b)

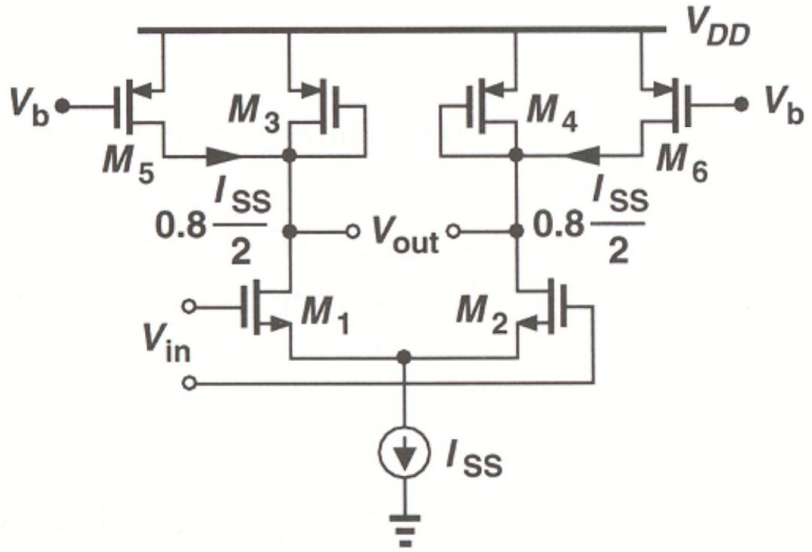
$$A_v = -g_{mN}(g_{mP}^{-1} \parallel r_{oN} \parallel r_{oP}) \approx -\frac{g_{mN}}{g_{mP}}$$

$$A_v = -g_{mN}(r_{oN} \parallel r_{oP})$$

# Boosting Gain — Adding Auxiliary Current Sources

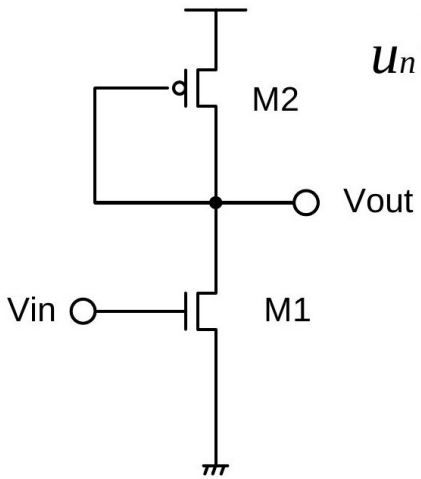
- Auxiliary current sources M5, M6 are added in parallel with the diode-connected loads M3, M4, each supplying a large fraction (e.g. 0.8·ISS/2) of the branch current
- Only the remaining fraction flows through the diode-connected device, lowering its effective gm and boosting the overall voltage gain

Addition of current sources to increase the voltage gain



# Gain Formula for the Basic Diode-Connected Load

- Equal branch currents (ID1 = ID2) with M1 (NMOS) and M2 (PMOS, diode-connected) in saturation give μn(W/L)1(VGS1−VTH1)² = μp(W/L)2(VGS2−VTH2)²
- Solving yields Av = −√[μn(W/L)1/(μp(W/L)2)] = −|VGS2−VTH2|/|VGS1−VTH1| — implying a substantial voltage-swing constraint



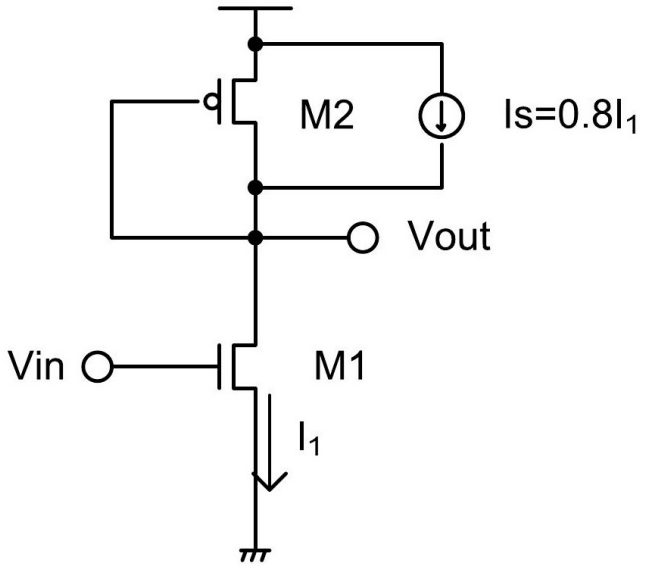

$$ID1 = ID2, \quad \therefore \quad u_n\left(\frac{W}{L}\right)_1 (V_{GS1} - V_{TH1})^2 = u_p\left(\frac{W}{L}\right)_2 (V_{GS2} - V_{TH2})^2$$

$$A_v = -\sqrt{\frac{u_n (W/L)_1}{u_p (W/L)_2}} = \frac{|V_{GS2} - V_{TH2}|}{(V_{GS1} - V_{TH1})}$$

This implies substantial voltage swing constraint.

# Quantifying the Gain Boost — Is = 0.8·I1

- With the auxiliary current source supplying 80% of the branch current, only 20% flows through the diode-connected device: u2S2VGT2² = (1/5)u1S1VGT1²
- The resulting gain Av = √(5u1S1/u2S2) = 5·(VGT2/VGT1) shows roughly a 5× gain improvement from diverting 80% of the current around the diode-connected load




$$u_2 S_2 V_{GT2}^{\ 2} = \frac{1}{5} u_1 S_1 V_{GT1}^{\ 2}$$

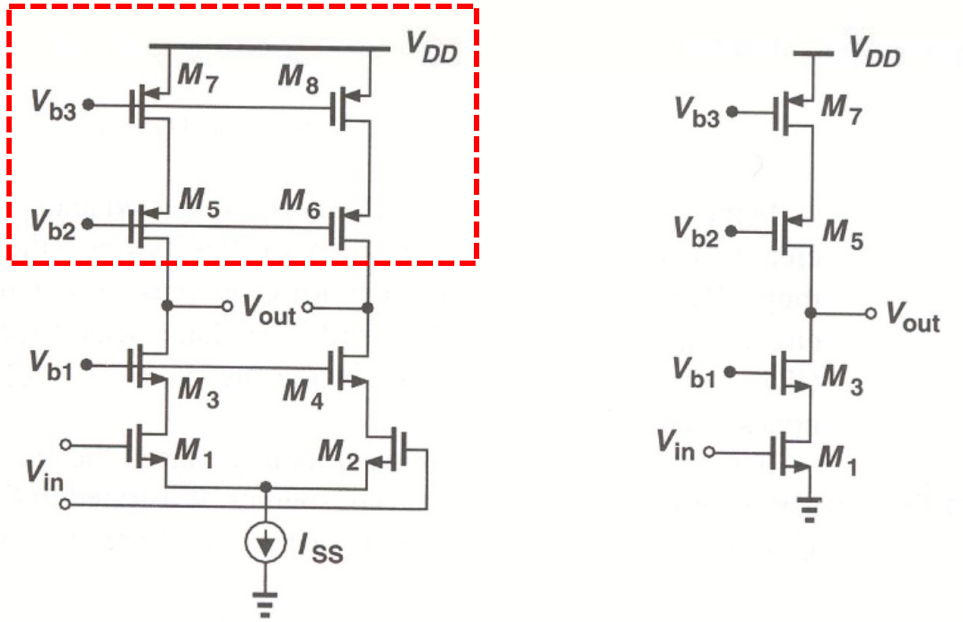
$$A_V = \sqrt{\frac{5u_1 S_1}{u_2 S_2}} = 5\frac{V_{GT2}}{V_{GT1}}$$

$$\frac{V_{GT2}}{V_{GT1}} = \frac{A_V}{5}$$

# Cascode Differential Pair

- Stacking cascode devices M3–M6 above the input pair and below the load boosts the output resistance on both the NMOS and PMOS sides, rather than relying only on lowering the load's gm
- The resulting gain Av ≈ gm1·[(gm3ro3ro1) ‖ (gm5ro5ro7)] combines the boosted resistances from both branches — a large increase over the simple diode-connected load

Cascode differential pair




$$A_v \approx g_{m1}[(g_{m3}r_{o3}r_{o1}) \| (g_{m5}r_{o5}r_{o7})]$$

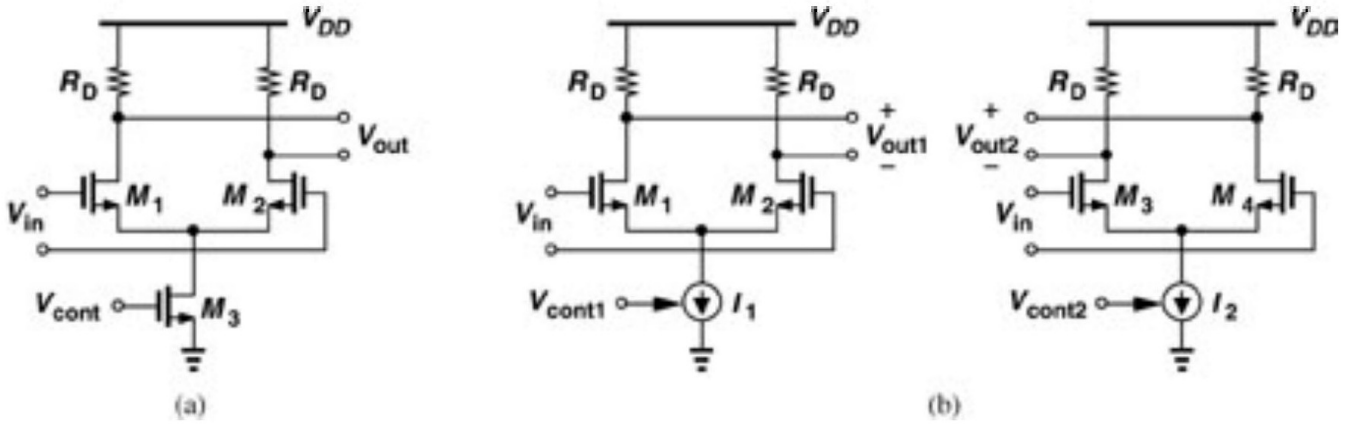
# VI

# Gilbert Cell

- Variable-gain amplifier concept
- Combining two differential pairs
- The cell as an analog multiplier

# Gilbert Cell — Building a Variable-Gain Amplifier

- Starting from a simple differential pair (a) whose tail current is steered by a control voltage, the sign of vout/vin flips depending on which side of a second differential pair (b) is enabled
- For Vcont1, vout/vin is negative; for Vcont2, vout/vin is positive — combining both control paths creates an amplifier with continuously variable, sign-switchable gain



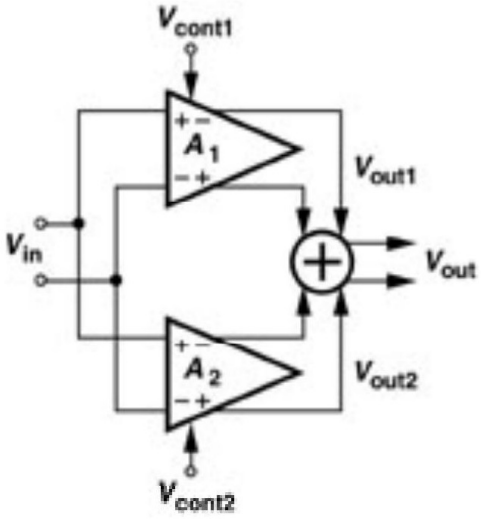

Firstly, start from a variable gain amplifier (a)
For Vcont1, vout/vin is negative
For Vcont2, vout/vin is positive

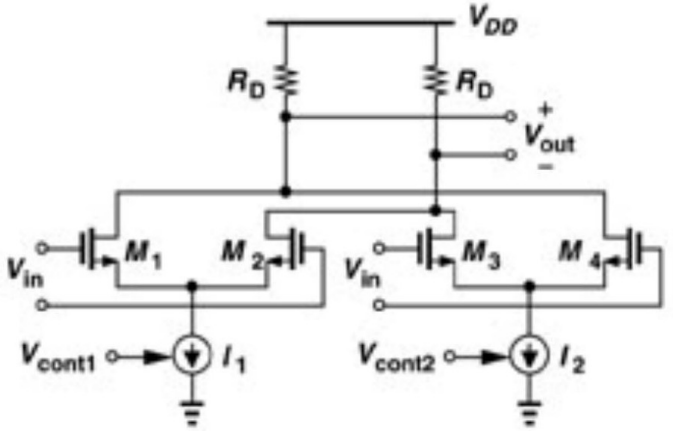
Combining Vcont1 and Vcont2, we can make a variable gain amp with both positive and negative gain

# Combining Two Differential Pairs at a Shared Load

- Two variable-gain sub-amplifiers, one controlled by Vcont1 and one by Vcont2, share the same input Vin and are summed onto a common RD load pair
- Vout1 = (ID1−ID2)RD and Vout2 = (ID4−ID3)RD combine to Vout1+Vout2 = (ID1+ID4−ID2−ID3)RD, the key step toward the full Gilbert-cell topology

How can we add vout1 for vcont1 and vout2 for vcont2?

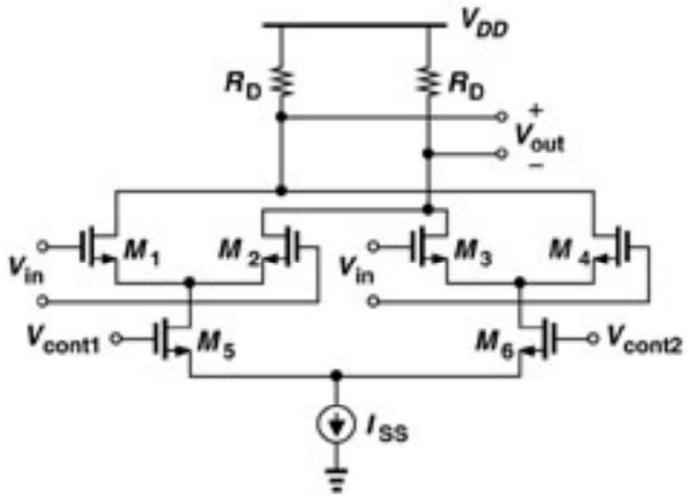


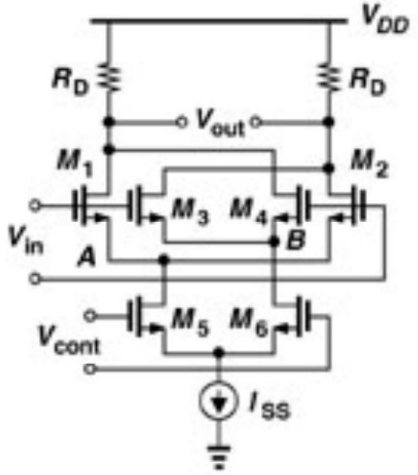




Vout1=(ID1-ID2)*RD, Vout2=(ID4-ID3)RD
Vout1+Vout2=(ID1+ID4-ID2-ID3)RD

# Complete Gilbert Cell Topology

- The full cell uses six transistors: an upper switching quad (M1–M4) driven by Vin, cross-coupled so their outputs add, and a lower differential pair (M5, M6) driven by Vcont that steers current between the two halves of the quad
- The overall result is VOUT = k·Vin·Vcont — the output is proportional to the product of the two input signals




$$V_{OUT} = kV_{in}V_{cont}$$

# Ex. 4.8 — Why the Gilbert Cell Multiplies

- From the variable-gain-amplifier view, Vout = Vin·f(Vcont) for some function f of the control voltage
- Expanding f(Vcont) in a Taylor series and keeping only the linear term near the operating point gives Vout ≈ k·Vin·Vcont — the defining behavior of an analog multiplier

Example4.8) Why does the Gilbert cell operate as an analog multiplier?

Vout=Vin*f(Vcont)

if we expand f(Vcont) in a Taylor series

Vout=Vin*kVCont